Independent reproduction check

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

Teach the evidence-to-conclusion chain for Independent reproduction check

Resolve metric and scoring disagreements from artifacts without inventing a successful reproduction.

Keep one habit: Model the main case → reveal the picture gradually → probe limits → assess independent reasoning → plan a changed-case return.

MAIN LESSON · SECTION 1

Prepare the connected research investigation

> Keep modelling, shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of independent reproduction check

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Run the investigation, with workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Opening retrieval

Without opening the old lesson, explain one key idea from Peer review and response. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of peer review and response

Reference example, not the only acceptable wording: Teacher explanation: Peer review challenges **evidence**, method and clarity. Responses should explain changes with artifacts rather than merely defend the original wording.

For each comment record concern, response, change location and new **evidence** if needed. If no new **evidence** exists, narrow the claim instead of promising an unperformed test.

Without opening the old lesson, explain one key idea from Writing the methods section. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of writing the methods section

Reference example, not the only acceptable wording: Teacher explanation: A methods section should let a reader reconstruct the design and computation. Detail should explain choices that affect **evidence**.

Report question, units, sampling, permissions, data roles, baseline, procedure, metrics, uncertainty assumptions, missing/exclusion handling and versions. Distinguish prespecified and later changes.

Without opening the old lesson, explain one key idea from Robustness across conditions. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of robustness across conditions

Reference example, not the only acceptable wording: Teacher explanation: Robustness checks examine a frozen procedure under declared varied conditions. Performance differences do not by themselves identify cause.

Keep baseline, metric and eligibility aligned across conditions. Report condition counts and failures separately; new conditions may extend **evidence** but cannot guarantee all conditions.

Pictured end goal

Explain independent reproduction check using the records and boundaries of this fictional investigation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Peer review and response

Writing the methods section

Robustness across conditions

Compact alternative: two 45-minute sessions for the core explanation and W1–W5 only. The main plan uses three sessions; schedule further time for the complete investigation, project, visual journey and any code examples. Independent reproduction checks whether another operator can use the recorded artifacts to recover the stated computation without hidden help.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the **mechanism** and terminology of independent reproduction check

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Model the whole case and check each inference

> Keep the independent answer private until after the first attempt.

Definitions and complete contract

Independent reproduction checks whether another operator can use the recorded artifacts to recover the stated computation without hidden help.

Provide manifest, fixture data, code, environment and expected metric definition. Record assistance and mismatches; reproduction is not new-data replication.

Reproduction item	Complete supplied exercise contract
Main metric conflict	Paper names mean squared error; supplied script averages absolute errors
Main numerical values	Residuals and reported scores not supplied
Added distinct fixture	Predictions 1, 3; targets 1, 1; residuals 0, 2
Separate picture	Original score 5/6; rerun 4/6; only Q3 scoring differs
Picture generation	No new model outputs generated
Changed archive	Q3 saved file missing
Independent case	Different validation split used because split IDs absent

Reproduction item	Complete supplied exercise contract
Evidence limit	No full-study independent reproduction or new-data replication performed by the demonstration

Main evidence ledger and checked key

Main metric audit	Supported statement
Named metric	MSE averages squared residuals
Implemented metric	MAE averages absolute residuals
Contract mismatch	Different computation; inspect/correct script or report with provenance
Actual original discrepancy size	Unknown; no main residual values supplied
Distinct fixture residuals	0 and 2
Fixture MAE	$(0+2)/2 = 1$
Fixture MSE	$(0^2+2^2)/2 = 2$
Fixture purpose	Demonstrates possible difference, not the missing original run
After correction	Preserve old artifacts/results and record a new documented rerun

Main MSE/MAE mismatch is a metric-contract problem: preserve original script/results and document any correction/rerun. Numerical original discrepancy is unknown. Distinct fixture residuals zero/two gives MAE one, MSE two. Picture original 5/6 ≈ 83.333%, rerun 4/6 ≈ 66.667%, disagreement Q3; inspect saved output, label and rubric without selecting a favored score. Missing Q3 file leaves exact verification unresolved.

Independent different validation split due to absent IDs means materials do not match; supply exact split membership/grouping and versions, preserve both attempts and record assistance/mismatches. Do not claim exact reproduction or new-data replication from a changed split or local fixture.

Independent reproduction check — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Operator gets different MSE because the supplied script averages absolute errors while paper names squared error.	Provide manifest, fixture data, code, environment and expected metric definition. Record assistance and mismatches; reproduction is not new-data replication.	Metric contract mismatch explains a candidate discrepancy. Correct/report script and rerun; preserve original artifact versions and results.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: Metric contract mismatch explains a candidate discrepancy. Correct/report script and rerun; preserve original artifact versions and results.

Qualification: A successful reproduction repeats the original **evidence** and may preserve its original design limitations.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Operator gets different MSE because the supplied script averages absolute errors while paper names squared error.

Metric contract mismatch explains a candidate discrepancy. Correct/report script and rerun; preserve original artifact versions and results.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Boundary and misconception probes

Reproduction audit	Checked interpretation
Same metric name, different formula	Compare implementations, not just labels
MAE and MSE differ in a fixture	Demonstrates metric contract; not proof of every original mismatch
Original and rerun scores disagree	Inspect raw records and scoring rule
Choose higher result	Does not resolve incompatible judgments
Missing Q3 file	Mark exact check unresolved; do not fabricate evidence
Different validation split	Not the same evaluation materials
Other operator received help	Record assistance rather than call it unaided
Same-data rerun	Not new-data replication or guaranteed future behavior

Separate pictured facts and full reasoning

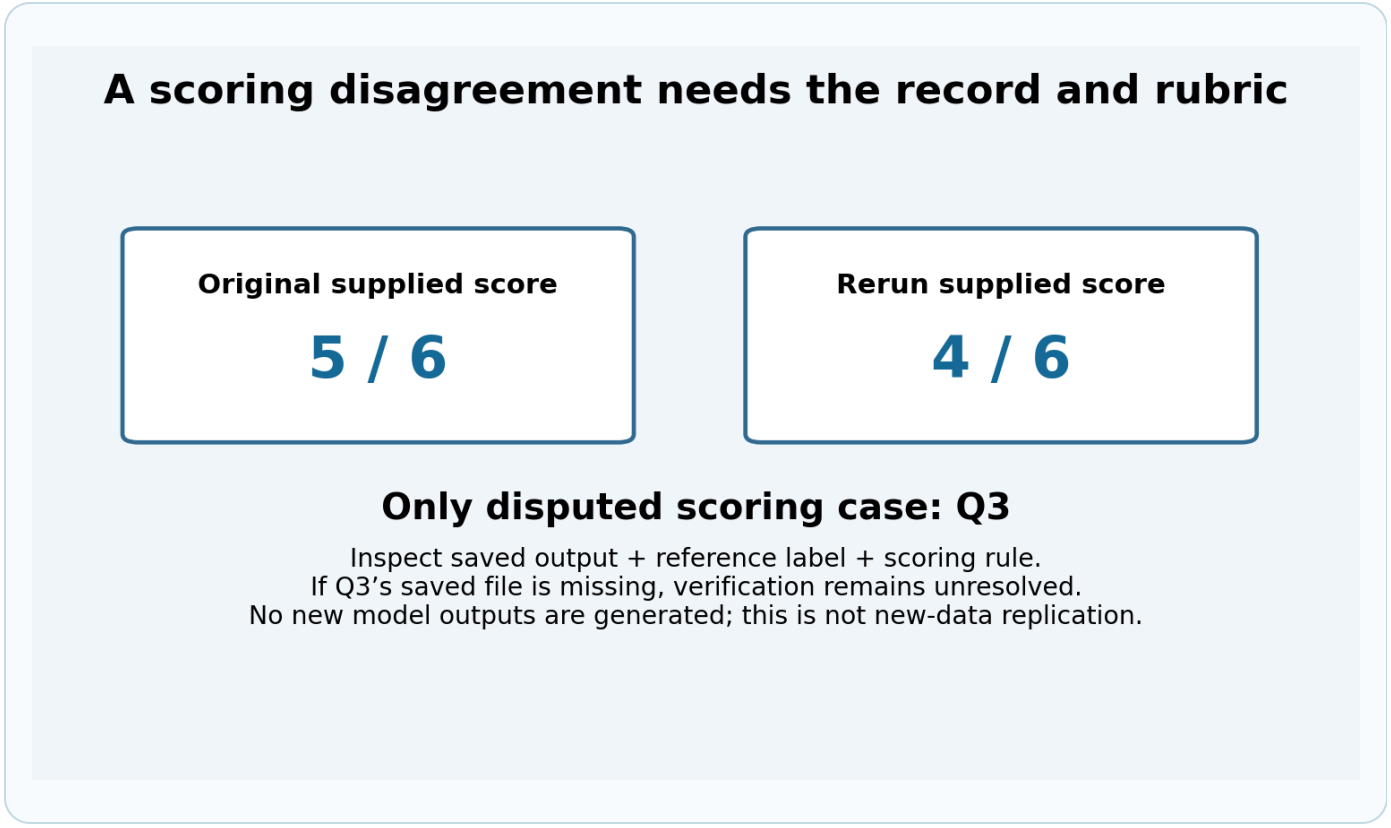
An independent rerun with a disagreement. A second team follows the saved protocol on the same fixed data and obtains a different score.

Original correct count is 5/6. Rerun is 4/6. Case Q3 is the only scoring disagreement. No new model **output**s are generated.

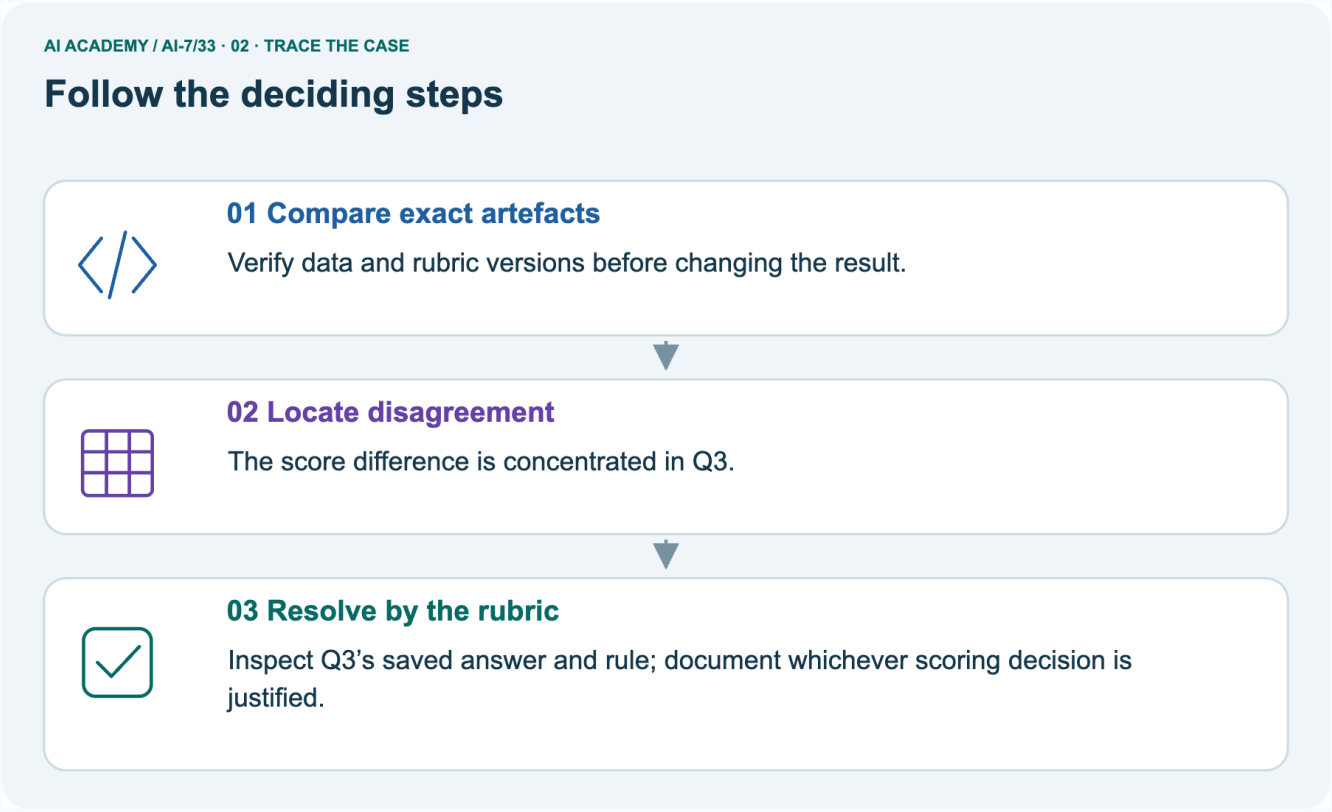
The separate picture’s original correctness count is five of six, while the supplied rerun is four of six. Only case Q3 has a scoring disagreement; no new model outputs are generated. Inspect Q3’s saved output, reference label and rubric before deciding how it should score. Do not simply take the higher count or average incompatible scoring judgments into a supposedly resolved result.

If Q3’s saved file is missing, exact verification of that case is unresolved; record the gap and any partial check honestly. The existing counts do not reveal the content needed for adjudication, and no program here can restore that missing evidence.

Separate scoring check	Supplied evidence or action
Original correctness	5/6 ≈ 83.333%
Rerun correctness	4/6 ≈ 66.667%
Descriptive difference	One case, about 16.667 percentage points
Disputed case	Q3 only
What to inspect	Saved output, reference label, rubric and scoring implementation
Adjudicated correct score	Not supplied; do not invent a winner
Q3 file missing	Exact verification unresolved; document missing artifact
Data role	Same saved outputs, not new-sample replication



Picture trace and changed-condition key



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Compare exact artefacts	Verify data and rubric versions before changing the result.
2. Locate disagreement	The score difference is concentrated in Q3.
3. Resolve by the rubric	Inspect Q3’s saved answer and rule; document whichever scoring decision is justified.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

An independent reproduction check can uncover interpretation or implementation differences. Do not average incompatible scores or simply choose the higher one.

This check concerns saved **output**s and scoring, not new-sample replication or future model behaviour.

Change: Q3’s saved file is missing from the archive.

Prediction: Mark exact reproduction incomplete until the required artefact is recovered or the limitation disclosed.

Reason: A missing record prevents checking the contested decision.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Independent operator	A person executing from the supplied record
Reproduction discrepancy	A mismatch needing documented investigation
Artifact manifest	A versioned inventory for the reported computation

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

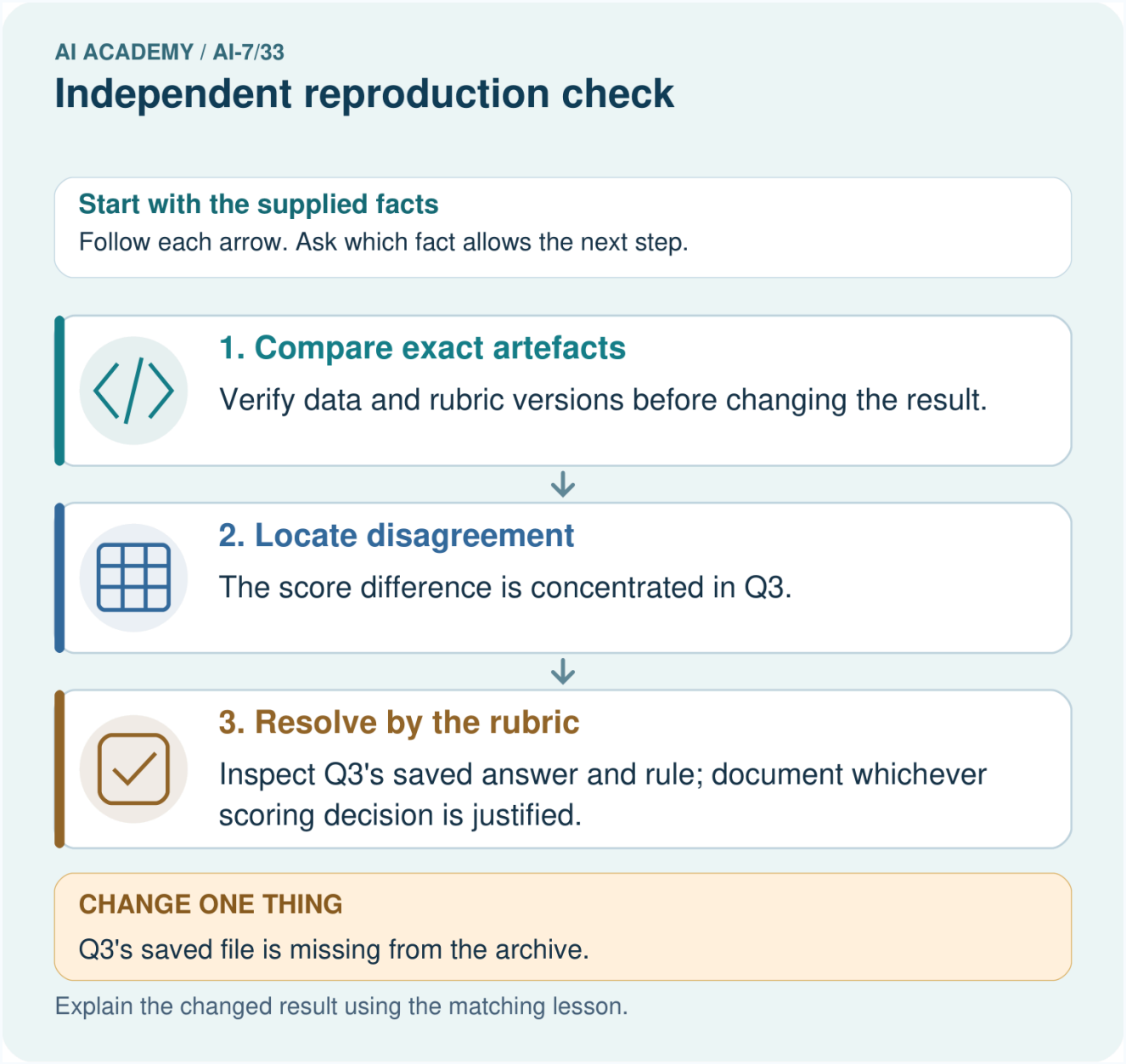
These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Independent reproduction check

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: Operator gets different MSE because the supplied script averages absolute errors while paper names squared error. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: Metric contract mismatch explains a candidate discrepancy. Correct/report script and rerun; preserve original artifact versions and results.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Independent reproduction check. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: A successful reproduction repeats the original **evidence** and may preserve its original design limitations.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Locate the Q3 scoring disagreement between 5/6 and 4/6. What should the teams inspect before choosing a score, and what limitation follows if the saved Q3 file is missing?

Verify data and rubric versions before changing the result. The score difference is concentrated in Q3. Inspect Q3's saved answer and rule; document whichever scoring decision is justified.

Changed case: Mark exact reproduction incomplete until the required artefact is recovered or the limitation disclosed. A missing record prevents checking the contested decision.

Changed-case reasoning: Mark exact reproduction incomplete until the required artefact is recovered or the limitation disclosed. A missing record prevents checking the contested decision.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use the worked example in Sections 3 and 6. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain See how it works and Words that unlock the idea. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main MSE/MAE mismatch is a metric-contract problem: preserve original script/results and document any correction/rerun. Numerical original discrepancy is unknown. Distinct fixture residuals zero/two gives MAE one, MSE two. Picture original 5/6 ≈ 83.333%, rerun 4/6 ≈ 66.667%, disagreement Q3; inspect saved output, label and rubric without selecting a favored score. Missing Q3 file leaves exact verification unresolved.

Independent different validation split due to absent IDs means materials do not match; supply exact split membership/grouping and versions, preserve both attempts and record assistance/mismatches. Do not claim exact reproduction or new-data replication from a changed split or local fixture.

Diagnose the current misconceptions

The worked result establishes every future case: A successful reproduction repeats the original **evidence** and may preserve its original design limitations.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Section 9 for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows	Next teaching action
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new research scenario

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Reproducer uses different validation split because no split IDs were provided. Diagnose.

Teacher explanation: Missing split record prevents exact reproduction; provide IDs and versions, then replay. New split scores are a different evaluation.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: Independent reproduction checks whether another operator can use the recorded artifacts to recover the stated computation without hidden help.

Provide manifest, fixture data, code, environment and expected metric definition. Record assistance and mismatches; reproduction is not new-data replication.

W2 Trace the worked case

Operator gets different MSE because the supplied script averages absolute errors while paper names squared error. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: Metric contract mismatch explains a candidate discrepancy. Correct/report script and rerun; preserve original artifact versions and results.

W3 Audit the limit

A report omits this qualification: A successful reproduction repeats the original **evidence** and may preserve its original design limitations.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: A successful reproduction repeats the original **evidence** and may preserve its original design limitations.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. A successful reproduction repeats the original **evidence** and may preserve its original design limitations.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: Metric contract mismatch explains a candidate discrepancy. Correct/report script and rerun; preserve original artifact versions and results.

Qualification: A successful reproduction repeats the original **evidence** and may preserve its original design limitations.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

Reproducer uses different validation split because no split IDs were provided. Diagnose.

Teacher explanation: Missing split record prevents exact reproduction; provide IDs and versions, then replay. New split scores are a different evaluation.

W5 Design a revealing follow-up

Independent assessment and fresh retry

W4: Missing split record prevents exact reproduction; provide IDs and versions, then replay. New split scores are a different evaluation. Require intermediate reasoning and the stated scope.

Fresh retry: Same artifact hash but different reported denominator. Expected reasoning: Inspect metric aggregation policy; integrity alone does not establish correct computation.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit research reasoning

> Use saved evidence and assumptions to choose the next question.

Project and next-week bridge

For Reproducible AI Research Study, use this week's independent task as a rehearsal for the final **evidence** pack: Reproducer uses different validation split because no split IDs were provided. Diagnose.

Keep actual **observation**s separate from expected results, preserve failures, compare a baseline and explain the limits.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Final project manuscript. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Delayed return and support record

Suggested return points in this level: ai-7/34, ai-7/36. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of independent reproduction check

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
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> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

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> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

Before class, solve the supported practice: Operator gets different MSE because the supplied script averages absolute errors while paper names squared error. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: Metric contract mismatch explains a candidate discrepancy. Correct/report script and rerun; preserve original artifact versions and results.

Check every learner before discussion: Use the worked case for Independent reproduction check. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: A successful reproduction repeats the original **evidence** and may preserve its original design limitations.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Q3’s saved file is missing from the archive. Require a **prediction** and a reason before comparing.

Expected result: Mark exact reproduction incomplete until the required artefact is recovered or the limitation disclosed. Reason: A missing record prevents checking the contested decision.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

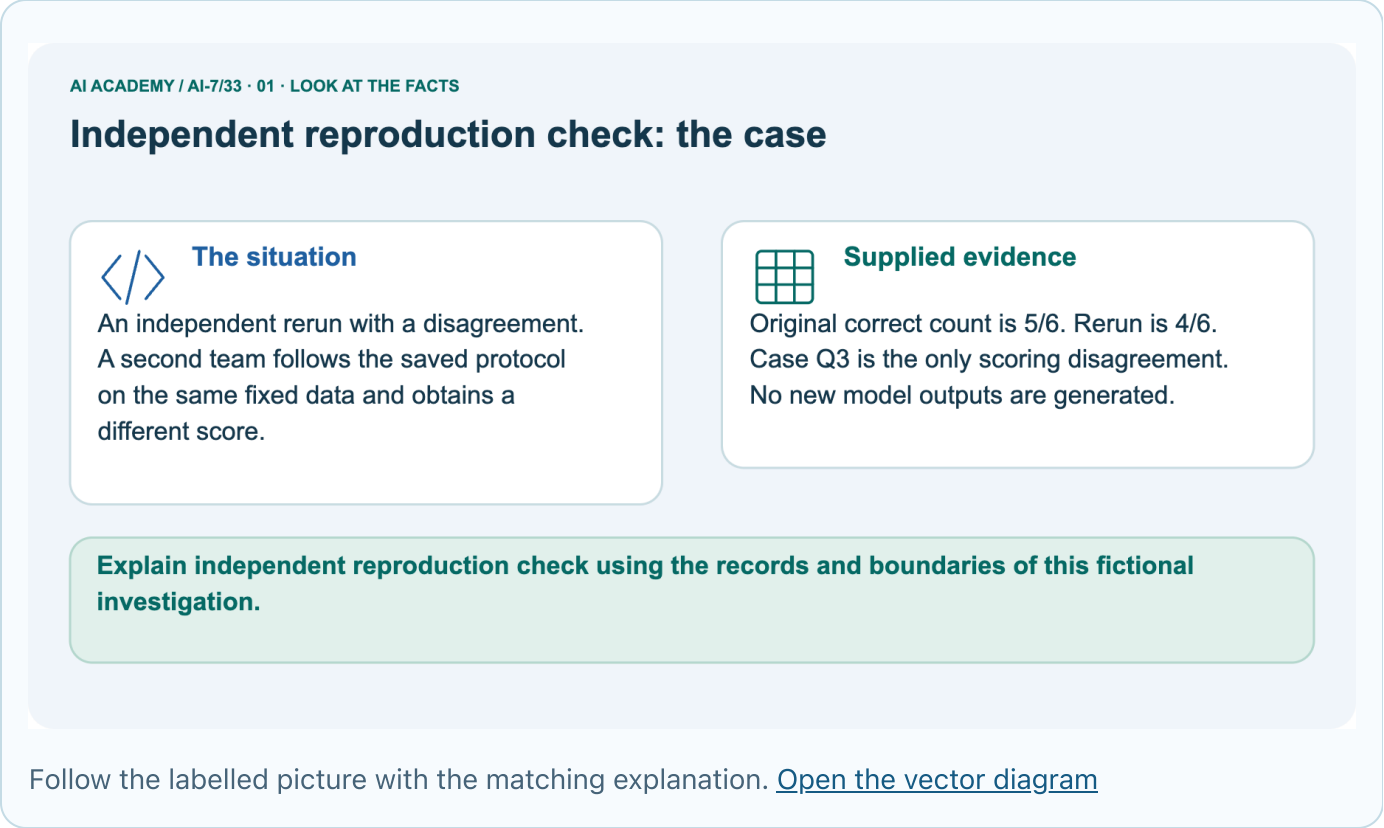
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Independent reproduction check: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

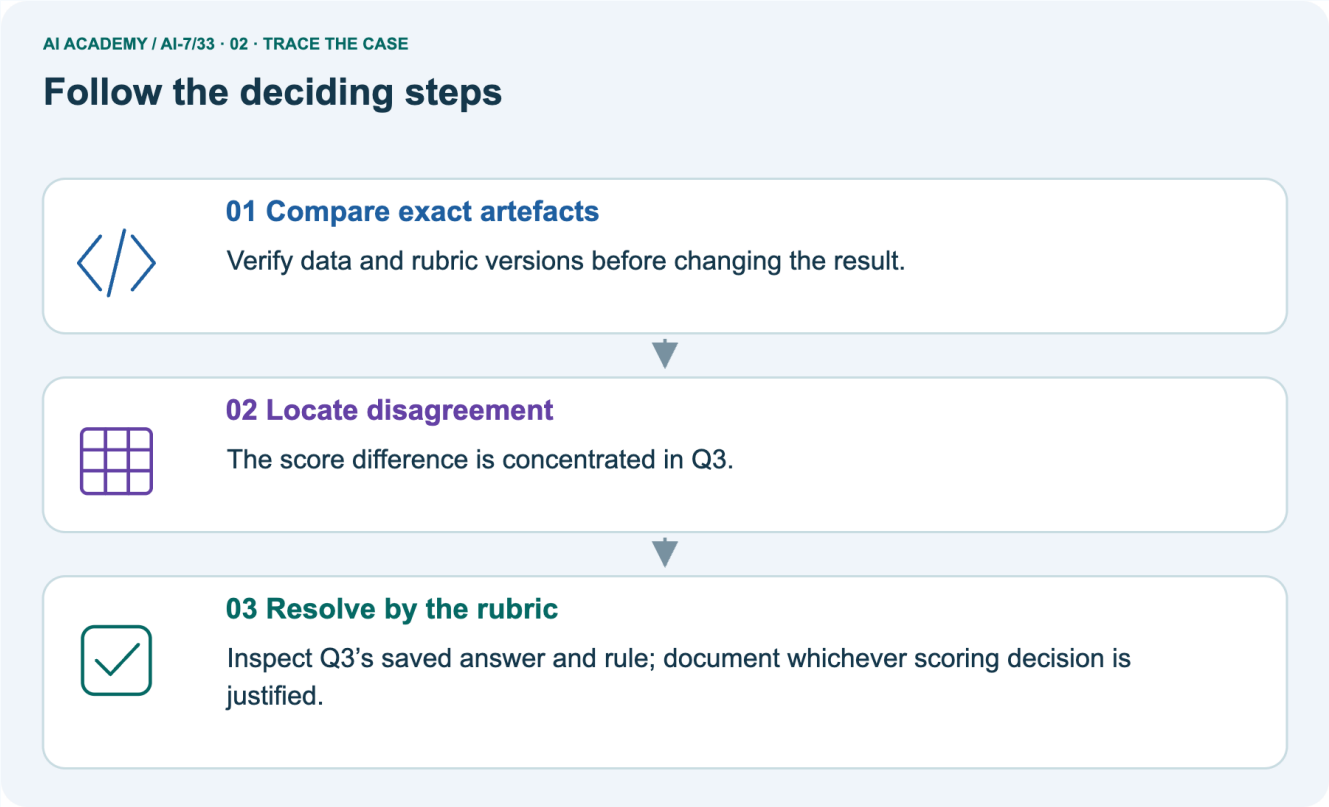
An independent rerun with a disagreement. A second team follows the saved protocol on the same fixed data and obtains a different score. Supplied facts: Original correct count is 5/6. Rerun is 4/6.

Case Q3 is the only scoring disagreement. No new model **output**s are generated.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



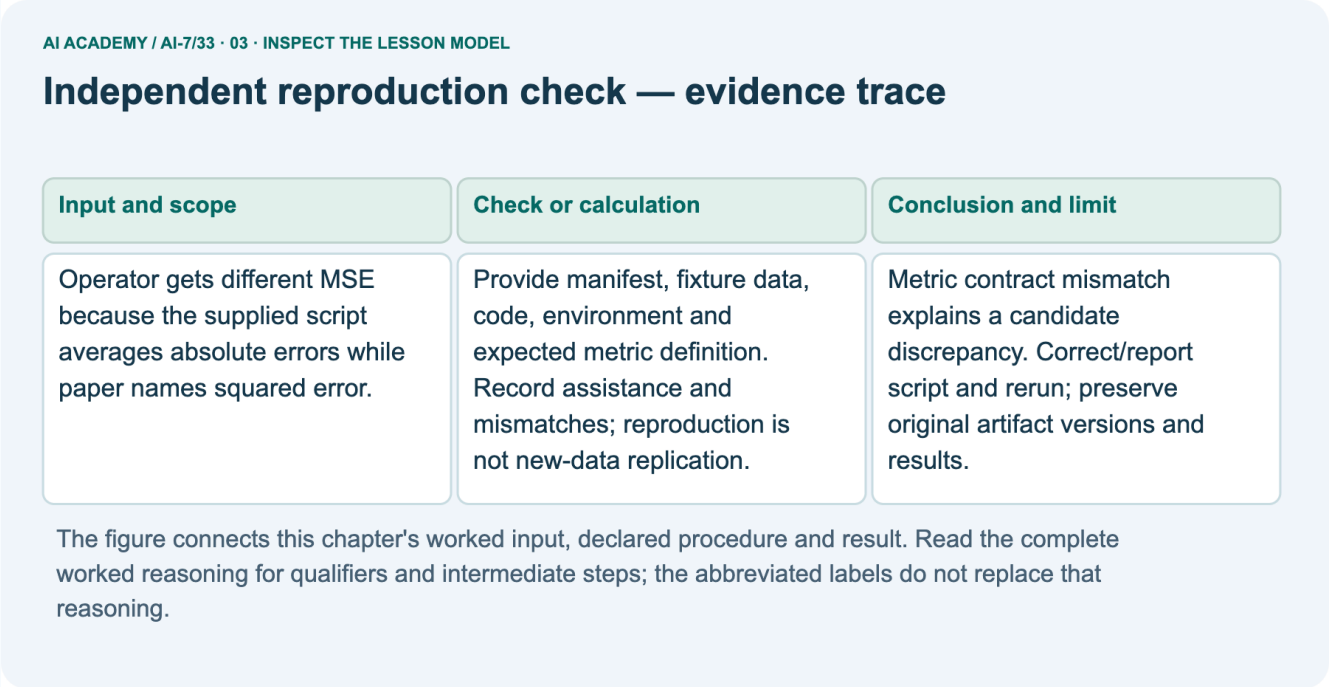
Follow the labelled picture with the matching explanation. Open the vector diagram

Compare exact artefacts: Verify data and rubric versions before changing the result. → Locate disagreement: The score difference is concentrated in Q3. → Resolve by the rubric: Inspect Q3's saved answer and rule; document whichever scoring decision is justified.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Independent reproduction check — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Independent reproduction check — **evidence** trace.

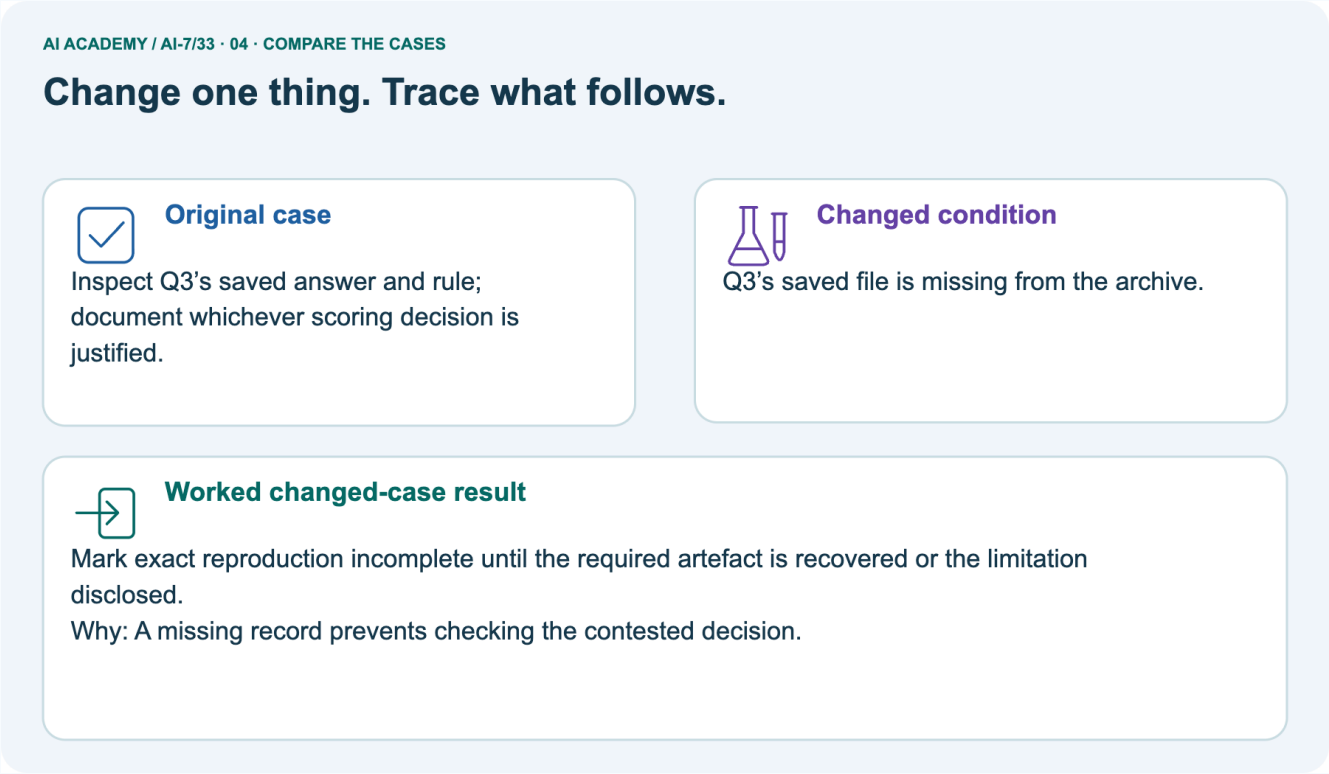
Operator gets different MSE because the supplied script averages absolute errors while paper names squared error. | Provide manifest, fixture data, code, environment and expected metric definition.

Record assistance and mismatches; reproduction is not new-data replication. | Metric contract mismatch explains a candidate discrepancy. Correct/report script and rerun; preserve original artifact versions and results.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Inspect Q3's saved answer and rule; document whichever scoring decision is justified. Changed condition: Q3's saved file is missing from the archive.

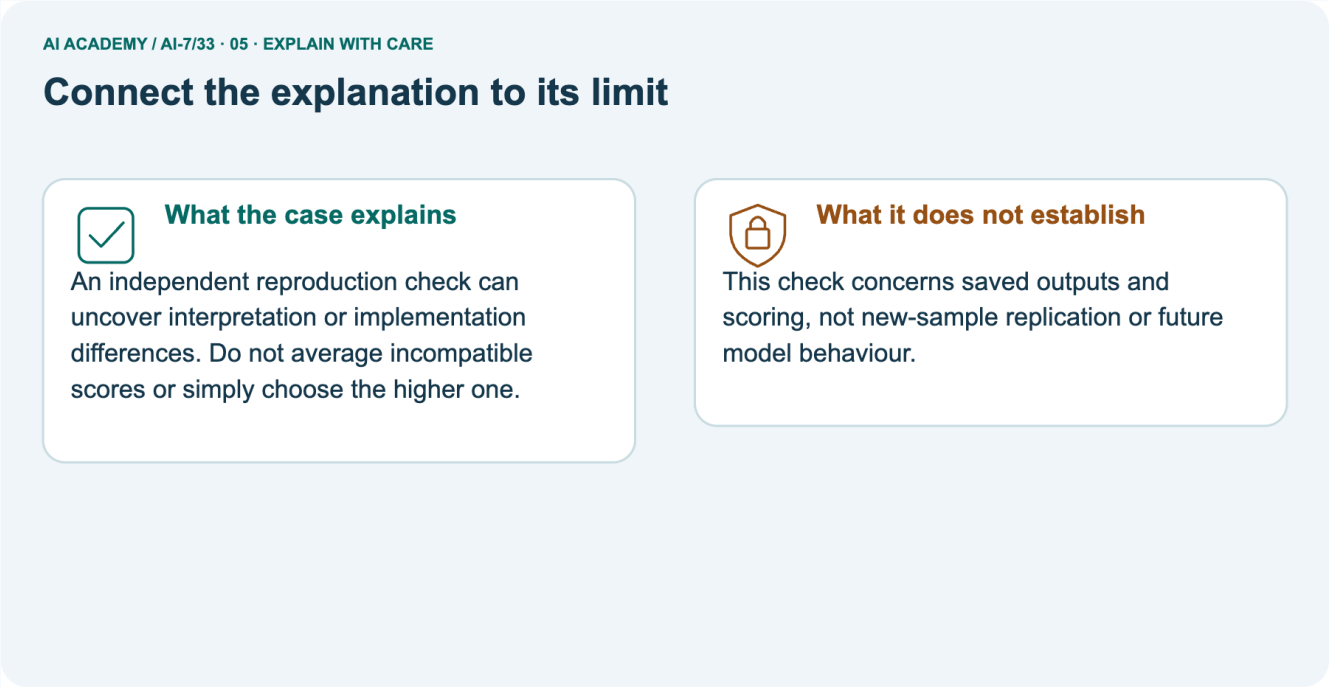
Expected result: Mark exact reproduction incomplete until the required artefact is recovered or the limitation disclosed. Reason: A missing record prevents checking the contested decision.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: A missing record prevents checking the contested decision.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

An independent reproduction check can uncover interpretation or implementation differences. Do not average incompatible scores or simply choose the higher one.

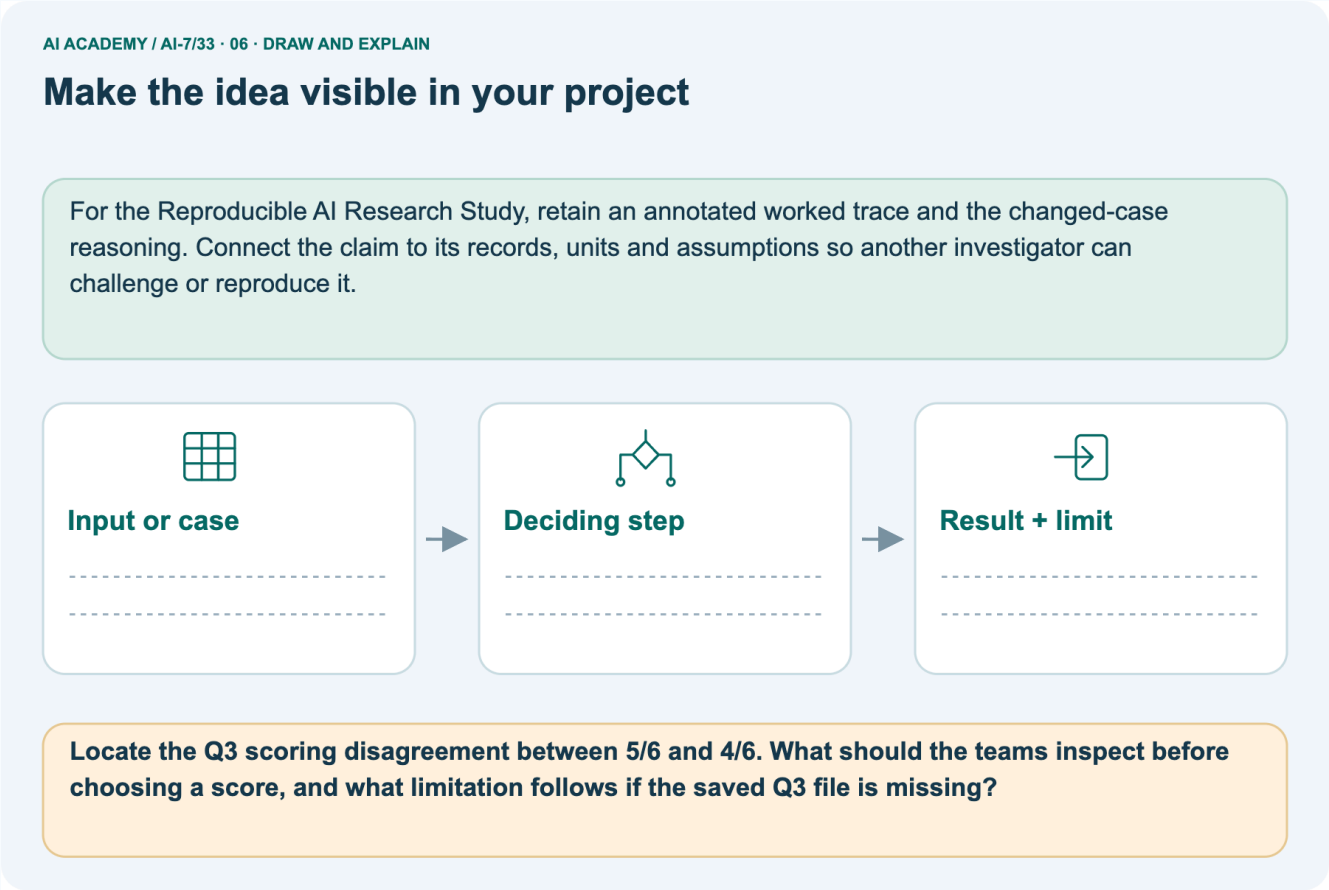
Limit: This check concerns saved **output**s and scoring, not new-sample replication or future model behaviour.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: This check concerns saved **output**s and scoring, not new-sample replication or future model behaviour.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Locate the Q3 scoring disagreement between 5/6 and 4/6.

What should the teams inspect before choosing a score, and what limitation follows if the saved Q3 file is missing?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Verify data and rubric versions before changing the result. The score difference is concentrated in Q3. Inspect Q3's saved answer and rule; document whichever scoring decision is justified.

Changed case: Mark exact reproduction incomplete until the required artefact is recovered or the limitation disclosed. A missing record prevents checking the contested decision.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Independent reproduction check: the case

02. Follow the deciding steps

03. Independent reproduction check — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.